

STOCKHOLM OBSERVATORIET, Sweden

WMO: 024840

Lat: 59.3415N

Lon: 18.0546E

Elev: 144

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.4	(c) 14.1	(d) 5.6	(e) 7.4	(f) 12.6	(g) 9.2	(h) 8.9	(i) 17.4	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.9	(c) 81.2	(d) 64.9	(e) 77.9	(f) 63.3	(g) 74.8	(h) 61.9	(i) 67.8	(j) 75.6	(k) 65.9	(l) 73.4	(m) 64.1	(n) 71.5	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	94.1	70.3	63.2	87.3	68.7	61.2	81.0	67.2	32.2	76.3	30.7	73.4	29.4	71.6	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	7.2	85.9	6.5	2.8	2.6	87.9	-1.2	89.5	-4.8	91.1	-9.5	93.2
			6.6	69.6	6.4	2.6	2.0	71.5	-1.7	73.0	-5.3	74.5	-9.9	76.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	29.4	30.0	34.9	43.9	53.3	60.5	66.4	64.2	56.0	45.8	38.6	33.1
	DBStd	14.36	7.76	7.30	7.25	6.30	6.43	5.67	5.35	4.72	5.44	5.90	6.14	7.59
	HDD50	2943	638	559	469	201	37	2	0	0	14	156	343	524
	HDD65	6940	1103	979	934	632	365	158	47	72	271	597	793	989
	CDD50	1639	0	0	0	19	140	316	507	439	194	24	0	0
	CDD65	161	0	0	0	0	3	22	89	45	2	0	0	0
	CDH74	838	0	0	0	0	26	130	517	163	2	0	0	0
	CDH80	141	0	0	0	0	1	13	107	20	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	21.7	1.5	1.1	1.0	1.2	1.3	2.0	2.8	2.5	2.2	2.0	2.0	1.9
	PrecMax	31.5	2.9	2.8	2.7	3.4	3.1	4.4	7.6	6.4	6.7	5.4	4.1	4.6
	PrecMin	14.8	0.2	0.2	0.0	0.2	0.3	0.1	0.7	0.1	0.5	0.4	0.6	0.4
	PrecStd	3.7	0.7	0.6	0.6	0.7	0.7	1.0	1.7	1.4	1.4	1.2	0.9	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	46.8	56.9	67.1	77.4	82.1	86.8	82.7	73.2	60.4	52.9	51.2
		MCWB	42.7	42.5	45.9	51.9	58.8	63.6	65.8	66.5	61.6	55.1	50.9	48.1
	2%	DB	43.0	43.4	51.4	62.3	72.6	77.7	83.2	78.2	69.3	57.7	50.4	46.9
		MCWB	40.8	39.5	43.0	49.7	57.0	61.1	65.5	64.6	59.3	53.8	48.4	44.3
	5%	DB	41.0	41.1	48.0	58.2	68.3	74.3	79.7	74.9	66.4	55.7	48.5	44.4
		MCWB	39.2	38.4	41.4	47.5	55.4	59.5	64.7	63.3	58.6	52.7	46.7	42.1
	10%	DB	39.0	39.1	44.5	54.4	64.4	70.8	76.4	72.1	63.7	53.8	46.6	42.3
		MCWB	37.3	37.1	39.3	45.0	53.3	58.2	63.2	62.2	57.1	51.2	45.0	40.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.7	44.0	46.8	53.6	61.7	65.7	70.6	70.7	62.9	56.9	51.5	48.2
		MCDB	45.1	46.2	54.7	63.7	73.3	77.4	79.1	77.9	70.0	58.9	52.4	50.8
	2%	WB	41.3	40.4	43.9	50.6	58.9	63.1	68.3	67.7	61.1	54.6	48.8	44.8
		MCDB	42.5	42.4	50.0	60.7	68.5	73.7	77.0	73.4	66.6	56.7	50.1	46.4
	5%	WB	39.4	38.7	41.9	48.1	56.4	61.1	66.6	65.6	59.5	53.3	46.9	42.5
		MCDB	40.7	40.6	46.9	57.1	65.6	70.7	75.0	71.4	64.4	55.4	48.2	43.9
	10%	WB	37.5	37.1	40.2	45.9	54.4	59.4	65.0	63.8	58.2	51.6	45.2	40.5
		MCDB	38.9	38.9	44.0	52.8	62.8	68.2	73.1	69.6	62.6	53.5	46.5	42.0
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.8	9.7	13.5	14.9	14.3	13.9	12.6	10.7	7.7	5.8	6.0
		MCDBR	7.4	8.0	14.3	19.1	21.1	20.0	18.8	16.3	13.8	8.9	7.2	7.1
		MCWBR	7.2	6.0	8.6	10.1	10.3	8.5	6.3	6.1	6.3	6.2	6.9	6.9
	5% WB	MCDBR	7.3	7.3	13.2	17.4	18.8	17.5	16.1	13.6	12.3	8.6	6.9	6.9
MCWBR		7.3	5.9	8.3	9.5	10.1	8.1	6.6	6.5	6.6	6.7	6.6	7.0	
Clear-Sky Solar Irradiance	taub	0.283	0.321	0.339	0.375	0.372	0.362	0.390	0.381	0.348	0.336	0.309	0.267	
	taud	2.219	2.287	2.355	2.324	2.397	2.457	2.400	2.420	2.497	2.417	2.254	2.140	
	Ebn at Noon	164	218	253	262	272	277	265	258	250	212	150	122	
	Edh at Noon	14	22	28	34	34	32	33	30	24	19	14	10	
All-Sky Solar Radiation	RadAvg	93	276	757	1272	1688	1875	1735	1370	886	387	123	63	
	RadStd	13	46	83	120	142	106	162	120	96	62	28	7	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-209	-261	N/A	N/A

Nomenclature: See separate page